Chapter 10

Periodontology Applied to Operative Dentistry

"Learning never exhausts the mind"

—Leonardo Da Vinci

This chapter presents an overview of the periodontal structures and their importance in restorative dentistry. Following a review of the anatomy and physiology of the periodontal tissues, periodontal contributory factors affecting restorative dentistry are discussed. The impact of periodontal therapy as part of the restorative treatment and the effect of the restorative treatment on the periodontium will also be presented.

The proximity of many dental restorations to the periodontium makes their relationship inseparable. To maintain a healthy periodontium and avoid chronic periodontal inflammation, restorations should be designed and performed properly. Dental restorations that re-establish function and esthetics, when required, and are supported by a healthy periodontium should be the goal of any restorative procedure.

When necessary, periodontal therapy to eliminate or control the etiologic factors that contribute to periodontal disease should be performed prior to restorative procedures. Timely periodontal therapy can avoid unesthetic results such as undesired position of the gingival margins. Healthy gingival tissues that frame the dentition are crucial to maintain the oral health and to enhance esthetics.

Basic Concepts of the Periodontium Relevant for Restorative Dentistry

The Gingiva

From an anatomic point of view, the gingiva represents the masticatory mucosa that is bound to the teeth and covers the alveolar processes. The gingiva involves the alveolar crest, the interdental bony septa and part of the alveolar process to the mucogingival junction in continuity to the lining of alveolar mucosa.[1]

The gingiva is described as free, attached and interdental gingiva, with the *free gingival groove* and *mucogingival junction* as main anatomical landmarks (Fig. 10.1).

Free Gingiva

The *free gingiva* is unattached and often characterized by the sulcus depth. It can be distinguished from the attached gingiva by the free gingival groove. The free gingival groove is present in only about one-third of the normal gingiva with similar occurrence between different genders and deciduous/permanent dentitions.[2]

Attached Gingiva

The *attached gingiva* is firmly attached to the periosteum of the alveolar process. The attached gingiva is clinically characterized by a firm and pink gingiva surrounding the dentition and separated from the mucosa by the mucogingival junction. In a series of radiographic studies, Ainamo and colleagues demonstrated the relative stable position of the mucogingival junction over time and suggested that the zone of attached gingiva increases with age due to teeth and alveolar process eruption.[3–5]

Clinical Notes

The widest band of attached gingiva is present on the buccal aspect of central and lateral incisors; the narrowest band is present at the buccal area of canines and first premolars.

Interdental Gingiva

The *interdental gingiva* consists of the gingival tissues that fill the embrasures below the interproximal contact points in anterior teeth. In posterior teeth where the interproximal contact points are broad, the interdental gingiva is formed from the buccal and lingual papillae bridged by the col.

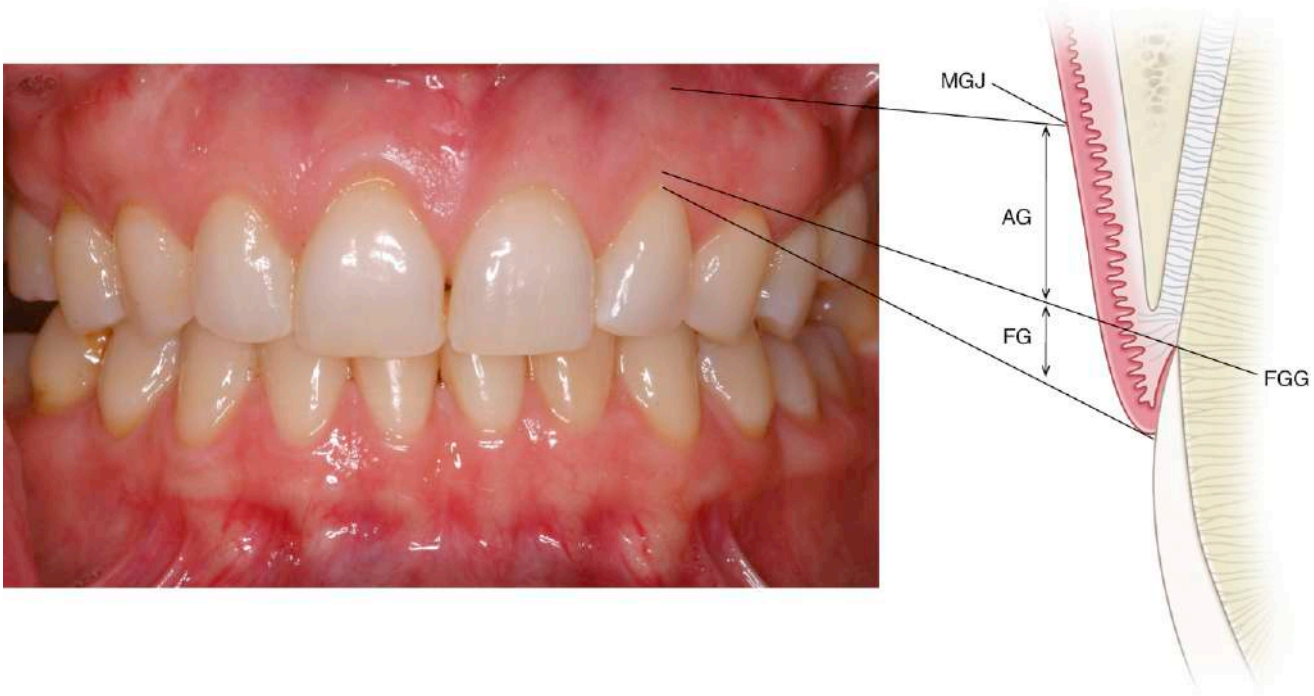


Figure 10.1 Free gingiva (*FG*) and attached gingiva (*AG*) are parts of the gingiva and limited by the anatomical landmarks mucogingival junction (*MGJ*) and free gingival groove (*FGG*).

The Cementum

The cementum is an avascular, multiunit, mineralized connective tissue with variable phenotypes (i.e. cellular, acellular, fibrillar, afibrillar) and functions. The principal function of the cementum is *tooth anchorage* (Fig. 10.2). Due to its dynamic restorative nature and adaptability, the cementum is important for the maintenance of the occlusal relationships and for the integrity of the root surface.[6]

The Alveolar Bone

The alveolar processes house the teeth and are composed of the outer cortical plates of compact bone, the trabecular bone and the alveolar bone proper (the bundle bone around teeth)[7] (Fig. 10.2). The cortical plates are built from bony lamellae and are thinner in the maxilla and thicker in the posterior mandible. The alveolar bone proper is located adjacent to the periodontal ligament.

The Periodontal Ligament

The periodontal ligament (PDL) occupies the space between the cementum and the alveolar bone (Fig. 10.2). The PDL is a multifunctional unit of connective tissue that contains numerous cells, fibres, rich vasculature and cellular components—osteoblasts, osteoclasts, fibroblasts, epithelial rests of Malassez, odontoblasts, cementoblasts, macrophages and undifferentiated mesenchymal cells. The extracellular matrix consists of collagen fibrils and other noncollagenous proteins[8] and pluripotent dynamic stem cells.[9]

From a functional point of view, the PDL participates in tooth anchorage, bone tissue development and homeostasis, nutrition–metabolic circulation and innervation.[7] Its width varies from 0.15 to 0.4 mm and is adapted to functional demands within this narrow range.[8]

The Gingival Display

In a healthy smile, the lip rises to the level of or slightly apical to the gingival margins of the maxillary anterior teeth, revealing 1–2 mm of gingiva. When more than 2 mm of gingival tissue is shown upon smiling, this is described as a '*gummy smile*' and considered unesthetic. Different aetiologic factors have been described in the literature for this excessive gingival display (Fig. 10.3).[10]

I. Excessive gingival display has been associated with *excessive maxillary bone growth*. In this case individuals may have longer lower third facial height, shorter upper lips and consequently more eruption of the maxillary teeth. A multidisciplinary approach including potentially combined orthodontic therapy and orthognathic surgery is indicated to move the maxillary complex apically and address this esthetic concern. Recently, surgical techniques for lip

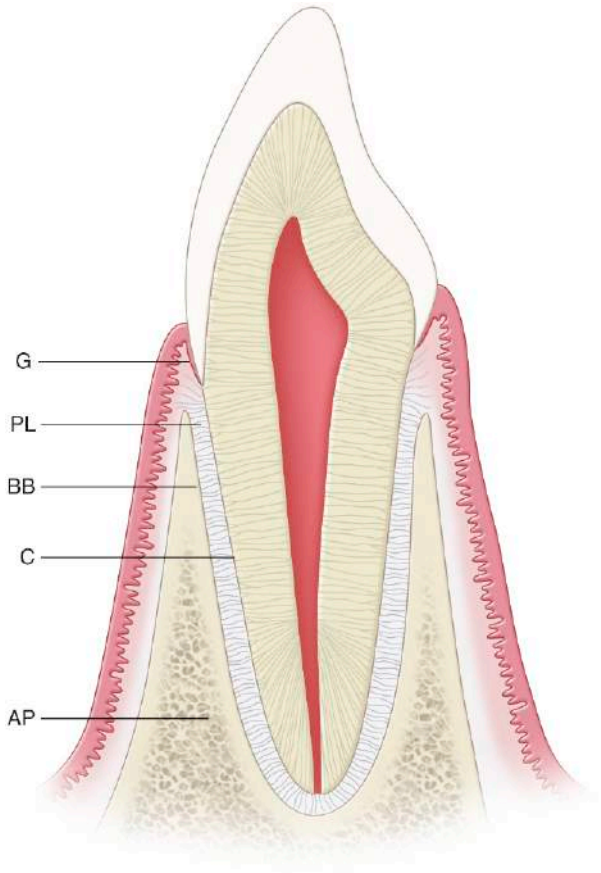


Figure 10.2 The periodontium surrounds and supports the teeth and is composed of gingiva (*G*), periodontal ligament (*PL*), bundle bone (*BB*), cementum (*C*) and alveolar process (*AP*).

repositioning have been described as alternatives to orthognathic surgery.[11,12] A comprehensive evaluation is always indicated to avoid frustration with unexpected results.

II. Another aetiologic factor for excessive gingival display is the delayed migration of the gingival margin commonly known as *altered passive eruption*. As teeth erupt, the gingival margin migrates apically to a level at or about 1 mm coronal to the cementoenamel junction (CEJ). The failure of gingival tissues to move apically results in excessive gingival display (Fig. 10.4).



Figure 10.3 'Gummy smile' showing more than 2 mm of gingival tissue during smile.
(Courtesy: Dr Tony Crivello.)



Figure 10.4 Example of altered passive eruption where the gingival margin failed to move apically and closer to the cementoenamel junction, resulting in excessive gingival display. (Courtesy: Dr André Ritter.)

Surgical intervention is usually beneficial in individuals with thick, fibrotic and noninflamed gingival tissue with a probing depth of 3–4 mm coronal to the CEJ. To ascertain the type of surgical intervention needed, the clinician is referred to the classification and treatment recommendations described by Coslet and colleagues.[13] Considerations in the decision between crown lengthening and gingivectomy are described later in this chapter.

Challenges in Periodontal Health Affecting Restorative Dentistry

Periodontal Disease

Periodontitis is an immune-inflammatory infection of the tooth-supporting structures and a major cause of tooth loss among adults. It has been recognized as one of the current most prevalent infectious diseases, affecting nearly 50% of the American population.[14] Associations with overall systemic conditions and numerous risk factors or indicators such as diabetes, smoking, age, genetics, immunosuppression and obesity have been identified as these may increase the severity and distribution of periodontal disease.[15]

A healthy periodontium is important to achieve long-term success with optimum comfort, function and esthetics for any restorative procedure. In the presence of active periodontal diseases, such as gingivitis or periodontitis, periodontal therapy should be performed prior to most restorative procedures in order to eliminate and/or control the aetiologic factors contributing to tissue damage.

Active periodontitis is diagnosed by clinical and radiographic evaluations. It is characterized clinically by:

I. deep probing depths,
II. inflammation (bleeding on probing or visual bleeding) and
III. subgingival calculus.

Radiographically, active periodontitis is characterized by *alveolar bone loss*. Signs of successful periodontal therapy are the resolution of gingival inflammation, which can be achieved with efficient oral hygiene habits and nonsurgical and/or surgical periodontal therapy. Administration of antibiotics is indicated in some situations.[16]

Similar to periodontitis, gingivitis is also diagnosed by gingival inflammation characterized by oedematous, spongy and red gingiva. Bleeding on probing is usually present as well. However, there is no loss of the periodontal attachment often characterized by radiographic bone loss. Different than periodontitis, gingivitis may be controlled with adequate oral hygiene and adult prophylaxis only. Periodontitis requires scaling and root planing to remove subgingival plaque and calculus deposits. Severe cases may require concomitant antibiotic therapy and periodontal surgery to re-establish a healthy periodontium. A healthy periodontium is important to establish a controlled and clean field for restorative procedures and to maintain the integrity and esthetics of restorative treatments.

The Importance of Maintenance Therapy

The end of active periodontal treatment, nonsurgical or surgical, is not the end of periodontal therapy. Only maintenance therapy can secure the long-term success of periodontal treatment. The effect of maintenance therapy is evident when comparing tooth loss, amount of bone loss, changes in attachment level and probing depth, tooth mobility and the progression of furcation involvement between patients with and without regular maintenance visits. It should be noted that tooth loss, while being the ultimate end point, may occur for reasons other than periodontal disease. Furthermore, the decision to extract a tooth is subjective and may depend on the preference and philosophy of the practitioner. Nevertheless, studies have shown that patients in periodic maintenance therapy preserve more teeth than those who fail to comply.[17,18] Changes in attachment level and probing depth are surrogate end points and are objective; however, the cut-off point determines the sensitivity of the parameters used. Longitudinal studies indicate that patients in a periodic maintenance therapy schedule may have less mean annual attachment loss than noncompliant patients.[19,20]

Although maintenance therapy can be very effective in stabilizing the progression of periodontal disease in most patients, a subset of patients (2%–4%) and sites may suffer from progressive attachment loss regardless of regular maintenance schedule.[21,22] As mentioned earlier, several risk factors such as smoking, stress and genetic background can influence disease progression.[23]

Gingival Biotypes

The gingival architecture depends on the type of gingival biotype. Generally, the periodontium has been described as having either of the two basic forms.[24,25]



Figure 10.5 Thin and scalloped biotype often shows transparency within the clinical gingival sulcus when a periodontal probe is inserted. It is often associated with tapered or triangular anatomic crown shapes with small interproximal contact areas.



Figure 10.6 Thick and flat biotype often shows more pronounced 'orange peel' surface characteristics of the attached gingiva. It is often associated with square anatomic crowns with broad interproximal contact areas.

Each biotype has its own distinct characteristics and the clinician must be mindful of these to achieve a stable dentogingival complex.

I. *Thin and scalloped biotype*: The thin and scalloped biotype frequently responds to gingival inflammation with gingival recession and is usually associated with tapered or triangular anatomic crown shapes with small interproximal contact areas (Fig. 10.5). Thin bone support with underlying dehiscence and/or fenestrations is a common finding in individuals with thin biotype.

II. *Thick and flat biotype*: On the other hand, thick and flat biotype often responds to gingival insults with deep pocket depth formation; and it is usually associated with square anatomic crowns with broad interproximal contact areas (Fig. 10.6). The stability of the bone crest and position of the free gingival margin are positively correlated to the thickness of the alveolar bone and gingival tissue. This was confirmed by Stetler and Bissada who showed that there was less inflammation and shrinkage when subgingival restorations were placed in individuals with thick gingival biotype.[26]

Furcation Involvement

The furcation forms with the presence of two or more roots in a tooth and is therefore most common in molars. It is

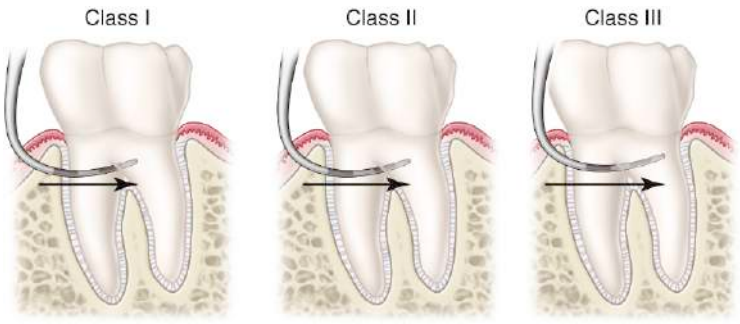


Figure 10.7 Furcation involvement classified into three different grades or classes by Hamp. Usually the furcation is examined with a specific probe such as the Nabers probe. The probe is placed against the tooth and moved in an apical direction as to try to find the entrance of the furcations.

assumed that furcations increase the surface area of teeth for periodontium attachment and thus provide additional periodontal support. However, this unique structure also renders neighbouring periodontium more susceptible to destruction.

Exposure of the furcation with its involvement may occur as a result of bone loss during progression of periodontal disease or of surgical treatment as part of the restorative therapy. The diagnosis of furcation involvement is crucial because it is related to tooth prognosis. The degree of furcation involvement dictates the treatment for the tooth, as there is an association between the severity of furcation involvement and tooth longevity.[27] For proper diagnosis, thorough clinical and radiographic examinations must be carried out.

Usually, the furcation is examined with a specific probe such as the ***Nabers probe*** (Fig. 10.7), a 12-mm curved probe with 3-mm markings. Placing the tip of the Nabers probe against the tooth and moving it in an apical direction aids in detection of furcation involvement. The depression of the furca may also be examined in a similar manner.

Unfortunately, even though special care is taken during clinical examinations, furcation involvement, especially in maxillary molars, is difficult to detect. The assessment of furcation areas tends to be inaccurate because of great variations in root anatomy. It is further complicated by limited access due to adjacent teeth or soft-tissue coverage. Radiographs provide an essential aid to clinical examinations; however, there are also limitations associated with the diagnosis of furcation involvement using radiographic images. The inherent two-dimensional presentation of conventional radiographic images limits the recognition of furcations.

Treatment options for furcation-involved teeth are generally based on the nature and extent of the furcation involvement. In general, treatment modalities are selected based on the classification of the furcation involvement. Several classification systems have been designed to determine the severity of furcation-involved teeth. The most commonly used is the ***Hamp classification***. It classifies furcation involvement into three different grades:

I. **Grade I:** Horizontal penetration of the Nabers probe less than 3 mm
II. **Grade II:** Horizontal penetration of the Nabers probe more than 3 mm, but not through the furcation
III. **Grade III:** Through penetration of the Nabers probe from buccal to lingual[28] (Fig. 10.8).

Other factors, such as size and divergence of roots, root trunk length, crown–root ratio and volume of remaining bone, are also important.

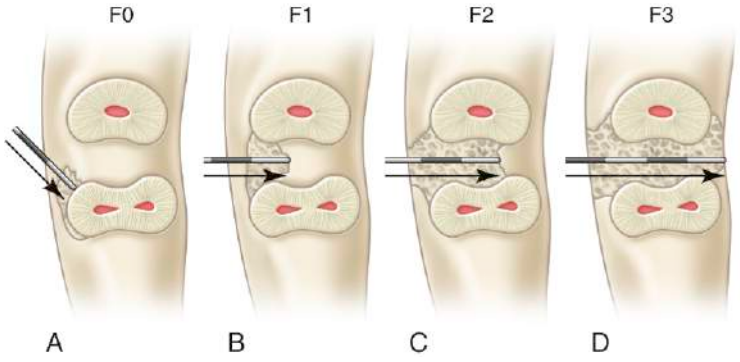


Figure 10.8 (A) *F0* indicates no furcation penetration with the Nabers probe. (B) *F1* grade or Class I indicates horizontal penetration of the Nabers probe up to 3 mm. (C) *F2* grade or Class II indicates horizontal penetration of the Nabers probe more than 3 mm, but not through the furcation. (D) *F3* grade or Class III indicates through penetration of the Nabers probe from buccal to lingual.

Management

I. Grade I furcation is generally treated with scaling and root planing. A clinical study evaluated the efficacy of scaling and root planing in Grade I furcation involvement and showed a 100% survival rate during a 5-year evaluation series.[28] Open flap debridement and furcation plasty may be indicated in some situations where prosthetic treatment is needed.

II. Grades II and III furcations are also treated initially with scaling and root planing. However, the need for surgical treatment, such as open flap debridement or guided tissue regeneration, may be indicated to improve tooth prognosis. An investigation on the effects of nonsurgical treatment on periodontally involved teeth with furcation involvement showed a 90.7% survival rate during a 5–12-year period. Out of the failures, 60% were on teeth with Grade III furcation and 30% on teeth with Grade II furcation demonstrates the limitations of nonsurgical therapy to treat severe furcation involvement.[29] A retrospective study looking at the long-term survival of maxillary furcation-involved teeth in patients who had received treatment and were on maintenance showed decent prognosis with only 12% of involved molars being lost in 24 years.[27]

Open Proximal Contacts

Interestingly, open proximal contacts do not have a direct effect on periodontal disease progression despite its reported indirect impact on the periodontium. In 1980, Hancock and colleagues analyzed 40 males with open proximal contacts between their teeth and could not demonstrate a significant correlation between open contacts, gingival inflammation and deep probing depths. On the other hand, open contacts were significantly related to food impaction, which demonstrated a significant correlation with higher gingival inflammation and deeper probing depths.[30] Jernberg and collaborators reported similar findings in regard to gingival inflammation and pocket depths. In their study, however, open proximal contacts demonstrated direct effect on the development of periodontal disease. A total of 104 adults were evaluated and open proximal contacts were found to be significantly correlated with deeper probing depths and higher clinical attachment loss.[31]

Clinical Notes

To reduce the influence of open proximal contacts, it is appropriate to restore (close) them as long as overhangs are not created. Adequate oral hygiene instructions should always be part of the therapy.

Enamel Pearls

Enamel pearls are ectopic globules of enamel adherent to the tooth found most often in furcation regions of maxillary second and third molars.[32]

They consist primarily of enamel, but dentin and even pulpal tissues can be found in some cases. The mean occurrence of enamel pearls in humans is 2.9% with a high occurrence of 9.7% among the Eskimo population.[32] Ectopic enamel such as enamel pearl is highly associated with attachment loss[33] and its removal is therefore indicated. Histologic observations demonstrate that enamel pearls have most of the structural features of enamel but with less orderly organization.[34] Its development is not well understood.

Cervical Enamel Projection

Cervical enamel projections (*CEPs*) *are ectopic deposits of enamel apical to the CEJ with a tapering form and extending towards or entering the furcation.*

Since connective tissue does not attach to enamel, CEPs have been demonstrated to be an aetiologic factor in furcation defects (Fig. 10.09). Bissada and Abdelmalek analyzed 1138 molars in Egyptian skulls and showed that CEPs were present in 8.6% of the cases. CEPs were present in approximately 50% of the cases where the furcation was involved.[35] The authors reported that CEPs were more frequently present in mandibular second molars followed by maxillary second molars. In another study, CEPs were present in 67.9% of the studied population (45.2% molars). Molars with furcation involvement had CEPs in 82.5% of the cases, whereas molars without furcation involvement had CEPs in 17.5% of the cases. The most affected tooth was the mandibular first molar followed by the maxillary first molar.[36]

The impact of CEPs on periodontal health depends on the extent of the projection. In 1964, Masters and Hoskins

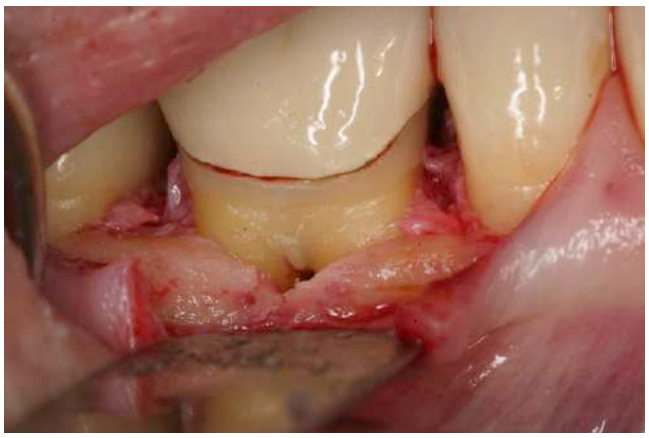


Figure 10.09 Cervical enamel projection present on the furcation of a mandibular molar. Note the furcation defect caused by lack of connective tissue attachment to the apically located enamel leading to increased bacterial colonization and apical migration in the furcation.
(Courtesy: Dr Tony Crivello.)

introduced a classification system for CEPs, which is still in use[37]:

I. Grade 1: Short but distinct change in the contour of CEJ extending towards the furcation
II. Grade 2: CEP approaches the furcation, without making contact with it
III. Grade 3: CEP extends into the furcation

Considering CEPs are a contributory factor to periodontal disease, especially in furcation areas, its removal or at least reduction is indicated. However, this should be done carefully to avoid irreversible damage to the tooth.

Cemental Tears

The cemental tear is a fragment of cementum that detaches from the tooth, usually after trauma, causing isolated bone loss (*vertical bone defects*) (Fig. 10.10).

It should be taken into consideration as a differential diagnosis for root fracture. Due to its low incidence, most studies available are case reports.

Treatment involves surgical debridement, removal of the cementum fragment and periodontal regeneration, if indicated. Several case reports demonstrate that the area can be successfully maintained after surgical treatment.[38,39]

Palato-radicular Grooves

The radicular or palatal groove is a developmental anomaly in which an infolding of the inner enamel epithelium and Hertwig epithelial root sheath creating a groove that passes from the cingulum of maxillary incisors apically onto the root.[40]

Palatal grooves may act as funnels, which facilitate the accumulation of biofilm in the depth of the groove. The area is inaccessible to both patients and clinicians and

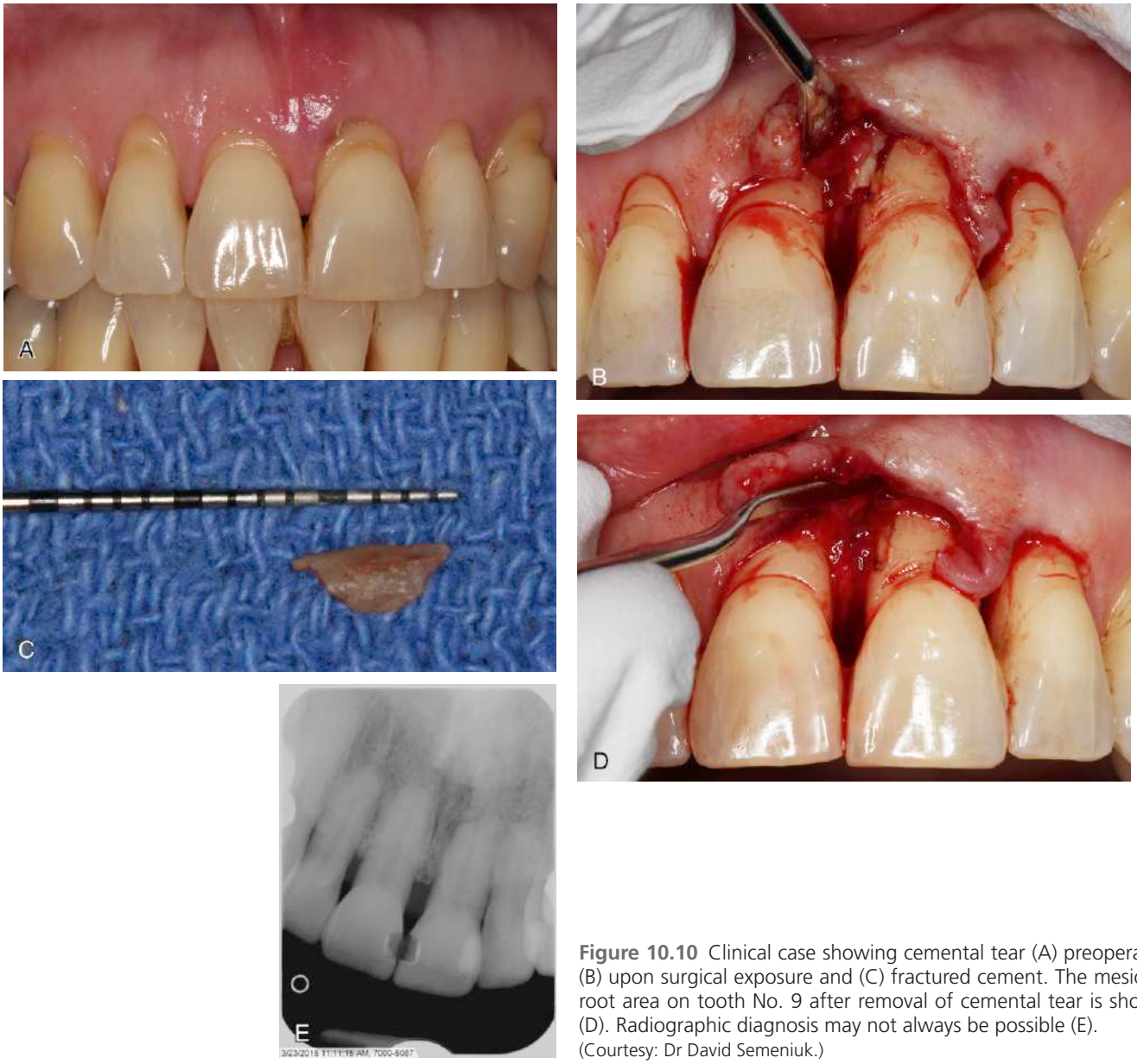


Figure 10.10 Clinical case showing cemental tear (A) preoperatively, (B) upon surgical exposure and (C) fractured cement. The mesiofacial root area on tooth No. 9 after removal of cemental tear is shown in (D). Radiographic diagnosis may not always be possible (E).
(Courtesy: Dr David Semeniuk.)

the prognosis for involved teeth, with diagnosed bone loss around the area, is poor. Hou and Tsai evaluated 404 maxillary incisors and reported that the palatal groove was present in 30.2% of lateral incisors, 5.9% of central incisors and 4% of both maxillary incisors.[41]

When indicated, treatment consists of odontoplasty to remove or reduce the groove. When odontoplasty is not possible, the groove may be restored with a biologically acceptable restorative material (see Materials, later in this chapter).

Tooth Position

Various parameters of tooth malposition have been correlated with periodontal disease.[42] The main influence of tooth malposition is the effect on plaque accumulation due to more difficult hygiene around malpositioned teeth. However, when oral hygiene is properly done, tooth malposition is not a significant factor causing plaque accumulation and consequent gingival inflammation.[43]

Occlusion

The role of traumatic occlusion in periodontal disease is extremely controversial. Consensus exist, however, that traumatic occlusion is not an initiating factor but can act as a contributory factor for periodontal disease. Nunn and Harrel evaluated the effects of traumatic occlusion on periodontal disease and concluded that occlusal discrepancy is an independent risk factor for periodontal disease.[44] Records from 24 years, showed that teeth with initial occlusal discrepancies demonstrate significantly worse prognosis, and deeper probing depths and mobility. Trauma from occlusion has also been linked to higher risk for furcation involvement.[45]

Based on the evidence suggesting that traumatic occlusion may influence the course of periodontal disease, minor occlusal adjustments with the purpose of achieving better periodontal treatment outcomes are recommended.[46]

The Biologic Width

The term *biologic width*, a genetically driven structure, refers to the combined vertical dimension of the junctional epithelium and the supra-alveolar connective tissue (Fig. 10.11).

This dento-gingival complex acts as a seal around the cervical portion of the tooth and has a self-restoration capacity. For instance, epithelial attachment mechanically separated from the tooth surface during periodontal probing or flossing reattaches to the original level in approximately 5 days.[47]

Clinical Notes

- In health, the epithelial attachment terminates at the apical end of the junctional epithelium.
- In diseased tissues, it terminates at the coronal aspect of the connective tissue or apical to the junctional epithelium.[48,49]

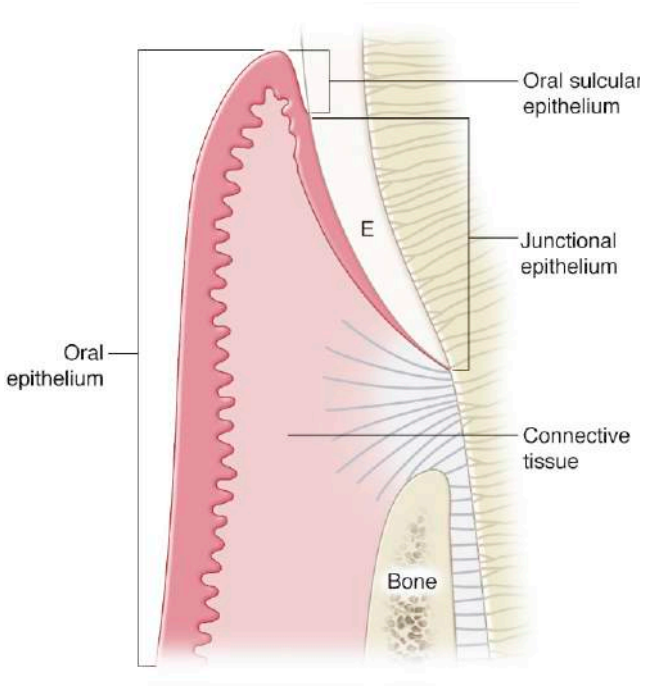


Figure 10.11 The biologic width combines the vertical dimension of the junctional epithelium and the supraalveolar connective tissue. The oral sulcular epithelium lines the clinical gingival sulcus, which is occlusal to the biological width. *E*, enamel.

Gargiulo and colleagues studied the dento-gingival dimensions in healthy human specimens and one of their observations was that the epithelial–connective tissue complex as a whole migrates apically during passive eruption. Moreover, they noted that during the different phases of development and passive eruption, the connective tissue zone had the most consistent dimension. The junctional epithelium was the most variable area.[50]

Years later, Vacek and colleagues focused on the implication of biologic width dimensions for restorative dentistry. They conducted another cadaveric study in pursuit of the minimum biologic dimensions that can be tolerated by the tissues. The authors confirmed the findings of the previous study in terms of the variability of the individual components. In addition, they recognized that the *biologic width was greater in posterior teeth* and the *junctional epithelium was significantly longer in teeth with restorations*.[51]

Clinical Notes

The mean biologic width was found to be **1.91 mm**, which consisted of:

- Junctional epithelium (1.14 mm)
- Connective tissue attachment (0.77 mm).

The mean sulcus depth was found to be 1.32 mm (Table 10.1; Fig. 10.11).

Table 10.1 Biologic width components and dimensions (mm and range)

	Gargiulo et al. 1961	Vacek and Gher 1994
Sulcus	0.69 (0.61–1.76)	1.32 (0.26–6.03)
Junctional epithelium	0.97 (0.71–1.35)	1.14 (0.32–3.27)
Connective tissue	1.07 (1.06–1.08)	0.77 (0.29–1.84)
Biologic width	2.04	1.91

Clinical Notes

Violating the biologic width by placing restorative margins within or apical to the junctional epithelium will lead to disturbance of the biologic seal and penetration of bacteria and their by-products leading to gingival inflammation, attachment loss and recession or pocket formation.

- In that event, the biologic width can only be restored by apical re-establishment of the supracrestal connective tissue.[52]

Periodontal Procedures Relevant to Restorative Dentistry

Crown Lengthening

A healthy periodontium is important to achieve long-term success and optimum comfort, function and esthetics for any restorative procedure. As mentioned earlier, the relationship between restorations and periodontium must be respected for a successful treatment outcome. Even if gingivitis or periodontitis are not present, one should not proceed with restorative treatment without further consideration of the future relationship between restoration and supporting tissues. In this regard, biologic width preservation is critical.

When violation of the biologic width—namely, placement of the restorative margin beyond the gingival sulcus and invading the junctional epithelium and connective tissue boundaries—is anticipated, crown lengthening of the tooth in question needs to be performed so the margin of the restoration is coronal to these biologic structures. As previously described, the mean depth of the sulcus is 1.32 mm; thus, placing a restoration beyond 1 mm into the sulcus is potentially disruptive of the biologic width.

Definition: *Crown lengthening is defined as the removal of bone tissue with concomitant removal or repositioning of the soft tissue around the tooth.*

The goal of this therapy is to increase the clinical crown and consequently preserve the biologic width. Typically, crown lengthening is needed to avoid impingement of the biologic width often occurring due to presence of subgingival caries or deep restorative margins. The procedure may also be recommended to correct esthetic unpleasant situations such as a 'gummy smile'. Failing to recognize the need for crown lengthening prior to restorative procedures often leads to invasion of biologic width leading to inflammation (gingivitis) and bone loss (periodontitis), pain associated with inflammatory remodeling of the periodontium, local soft- and hard-tissue defects, compromised esthetics and problematic retention of restoration in some instances.

Clinical Notes

- The biologic width is preserved when the restoration is placed allowing for approximately 2 mm for the connective tissue and junctional epithelium and 1 mm for the sulcus.
- Further, ideally an additional 0.5–1 mm should be added coronally to create a safe distance from the alveolar bone crest to the restorative margin.
- A 3–4-mm distance from gingival margin to the alveolar crest ensures periodontium healing after tooth preparation with quick re-establishment of the junctional epithelium and connective tissue integrity, which avoids continuous inflammation around the tooth.[53,54]

Current research and clinical observations show that there is variation in biologic width measurements among individuals and teeth, and that the *3-mm distance* is the minimum necessary to avoid periodontal breakdown around subgingival restorative margins.[55] Wagenberg and colleagues, in fact, have suggested to aim for a 5-mm distance from bone to the restorative margin when possible.[56]

The challenge is that periodontists have a tendency to remove less bone than needed due to the nature of their specialty (i.e. preservation of the periodontium).[57] Thus aiming for 5 mm will direct the surgeon to achieve the necessary removal of the minimum 3 mm. Other authors have also discussed that individual variations in biologic width dimensions will be accounted for if more than 3-mm distance is ensured after surgery.[58] Several studies have reported that subgingival margins of restorations, even without overhangs, have more plaque accumulation, gingival inflammation and development of periodontal pockets than supragingival ones.[59,60] A 26-year prospective study showed a 0.5-mm increased mean loss of attachment in patients with restorations with subgingival margins after 10 years of function.[61]

It is important to remember that the periodontium biotype plays a key role in the periodontal response. Normally if the periodontium is thin and scalloped, recession will follow the inflammatory process caused by the restoration margin. If the periodontium is thick and flat and thus soft tissue stability is more likely, there is a higher likelihood of a moat-like lesion around the tooth to appear on the alveolar bone, leading to increased bleeding, sensitivity on

probing and pocket formation.[53,62] Timely recommendation of crown lengthening to avoid such issues is a crucial step towards tissue health, predictable esthetics and patient satisfaction.

There are situations, however, where clinical *crown lengthening may not be indicated* and alternative treatment options should be explored. These include limitations caused by:

I. position of the furcation;
II. reduced attachment levels;
III. esthetic issues;
IV. teeth with significantly reduced periodontal support;
V. endodontically infected teeth that should not be treated prior to resolution of the periapical lesion;
VI. extensive need for osseous removal, which would compromise the stability of adjacent teeth and
VII. inadequate crown-to-root ratio after surgery.

For instance, a tooth with a short trunk may limit the amount of crown lengthening that can be performed as the procedure could expose the furcation and compromise tooth prognosis (Fig. 10.12). In that event, avoiding violation of biologic width may not be possible and the surgical procedure may not be performed or may be only partially performed, leading to a compromised restorative treatment. In general, all subgingival margins compromise and impact the periodontium. A higher degree of inflammation leading to bone loss is expected in these areas since proper crown lengthening was not performed.[57]

Another common scenario where biologic width is violated and difficult to manage is crown fractures extending below the gingival margin. Crown lengthening is often necessary for proper restoration of fractured teeth (Fig. 10.13). In the case of fracture of anterior teeth, crown lengthening surgery will significantly impact the tooth clinical and consequently esthetic display (as well as of the adjacent teeth) due to the removal of bone and subsequent apical repositioning of the gingival margin (Fig. 10.14). There are situations where orthodontic extrusion is indicated to minimally affect the gingival margin levels in the anterior sextant or esthetic area. Crown lengthening may follow the orthodontic extrusion. Camargo, in 2007, stated that forced tooth eruption via orthodontic extrusion is the technique of choice when clinical crown lengthening is necessary on isolated teeth in the esthetic zone.[63]

For optimal outcome, proper healing of the periodontal tissues after crown lengthening is necessary. The type of periodontium (thick and flat versus thin and scalloped) and the location (anterior versus posterior) generally dictate the extent of healing time. Direct restorations can be made as soon as there is initial tissue shrinkage if the margin is supragingival and haemostasis/trauma to the tissue is not anticipated. For indirect full-coverage restorations, clinical studies indicate that the *clinician should wait approximately 6–8 weeks before proceeding with final preparation and impression in posterior teeth*.[64]

In anterior esthetic cases, the periodontal biotype needs to be taken into consideration before proceeding with restorative treatment. In a thick and flat periodontium, the gingiva tends to bulk postoperatively prior to moving coronally and flatten as maturation occurs in the first 6–8 weeks. Further flattening may continue to occur

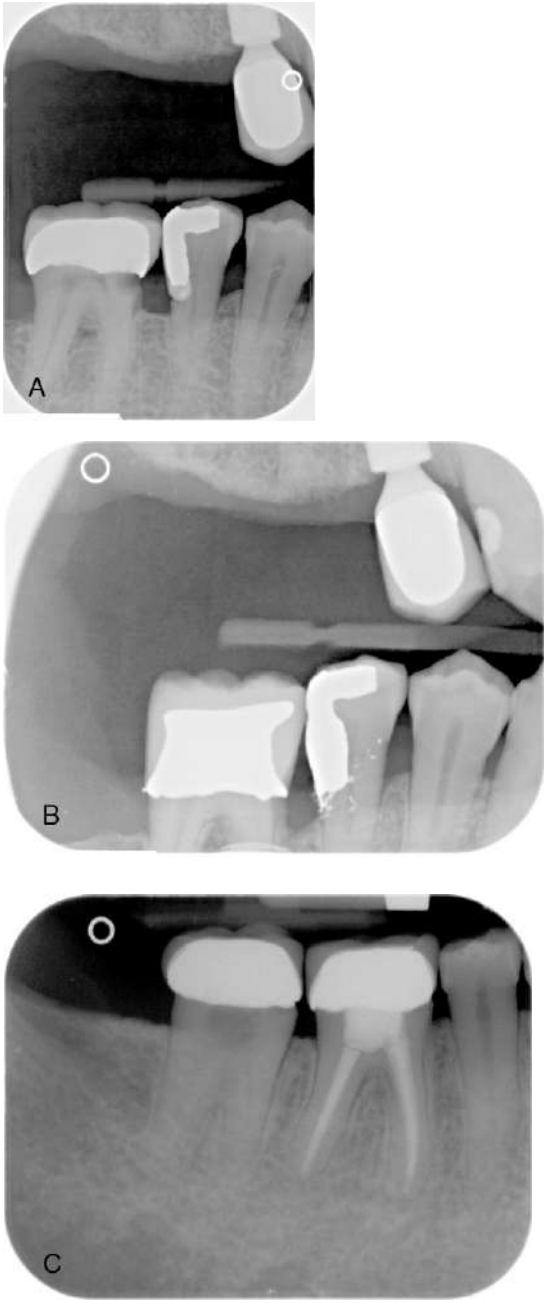


Figure 10.12 (A) Tooth No. 30 with secondary caries on the mesial surface leading to invasion of biologic width (beyond the clinical gingival sulcus). Note that tooth No. 29 has a restoration with deep and open margin violating the biologic width on the distal surface. (B) After compromised crown lengthening surgery and crown replacement. To avoid furcation exposure (note the short trunk of tooth No. 30), the amount of crown lengthening was limited. (C) Another clinical case showing an example of a long trunk on teeth No. 30 and No. 31, which would allow for a non-compromised crown lengthening surgery since furcation would likely not be at a risk of exposure.

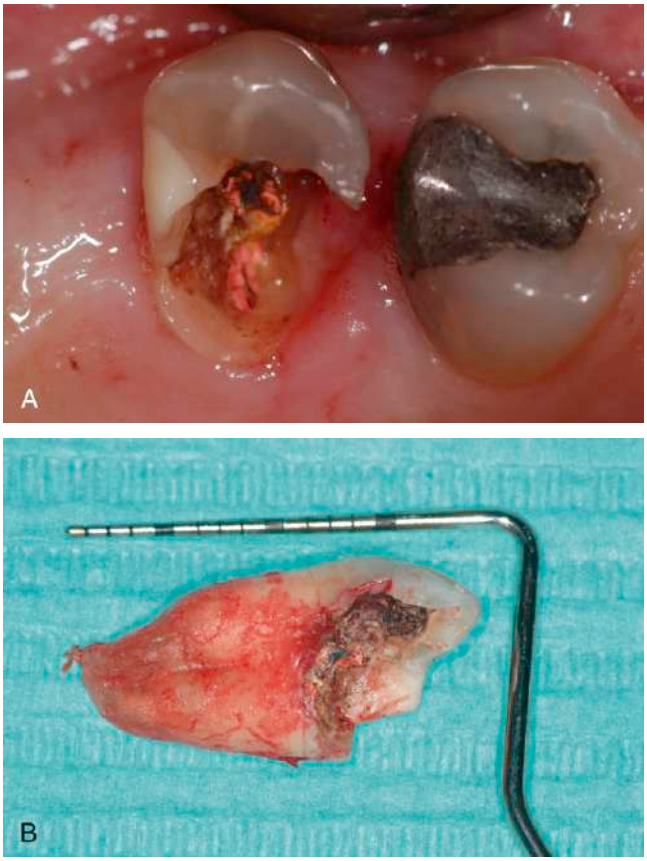


Figure 10.13 (A) Endodontically treated bicuspid presents with a large coronal subgingival fracture. (B) Poor crown–root ratio can be appreciated indicating that crown lengthening would not be recommended on this tooth to expose the fractured surface for restoration. Extraction was recommended.



Figure 10.14 Fractured of anterior tooth #7. Esthetics must be carefully considered when treatment planning the restoration of anterior fractured teeth. Crown lengthening may compromise the esthetic outcome and be contraindicated.
(Courtesy: Dr Eduard Epure.)

beyond the initial maturation phase. In a thin and scalloped periodontium, there is usually recession immediately after surgery and in some cases 'creeping attachment' occurs in the months after initial maturation. One study demonstrated that probing depths stabilized at 6 weeks but recession ranging from 2 to 4 mm continued to develop between 6 and 24 weeks.[65] Thus, in areas of esthetic concern, postponing final margin placement and restoration for 6–8 months may be desirable to ensure gingival margin stability. To better predict results, an accurate diagnosis of thick versus thin biotype should be done.[66]

Gingivectomy

Unlike crown lengthening, gingivectomy does not involve hard tissue but rather just gingival excess removal to expose the clinical crown.

For a successful gingivectomy procedure, careful treatment planning after detailed examination is crucial. Diagnosis and management of the etiology are critical for a positive outcome.

I. In the case of *gingival overgrowth*, the soft tissue should not rebound provided the etiology for overgrowth is addressed (e.g. triggering medications, local trauma) and good oral hygiene is exercised. Plaque control is critical for overgrowth control in cases involving drugs that may stimulate tissue reaction and where medication replacement is not possible.[67]
II. In the case of *altered passive eruption*, where there is excess of soft tissue due to altered apical migration of the gingiva, the tissue removed does not regrow provided absence of coronal positioning of the bone. When gingival tissue is removed to expose the clinical crown but the removal alters the dimension of the sulcus and junctional epithelium, the tissue regrows to establish at least approximately 1 mm of junctional epithelium and 1 mm of sulcus. However, this can be affected by genetically predetermined dimensions. Crown lengthening would be indicated to remove the bone and reposition the gingival margin apically.

Some of the clinical scenarios to consider gingivectomy versus crown lengthening are:

I. subgingival caries/access for proper restoration,
II. tooth fracture,
III. inadequate retention,
IV. altered passive eruption and
V. other esthetic concerns such as gingival overgrowth or defects.

i. Subgingival Caries/Access for Proper Restoration

When subgingival caries is present, at least 3 mm of sound tooth structure are needed from the margin of the final cavity preparation (not the caries lesion) to the alveolar bone crest (Fig. 10.15); otherwise crown lengthening is recommended.

It is important to emphasize that open flap and bone contour correction is warranted if periodontal breakdown

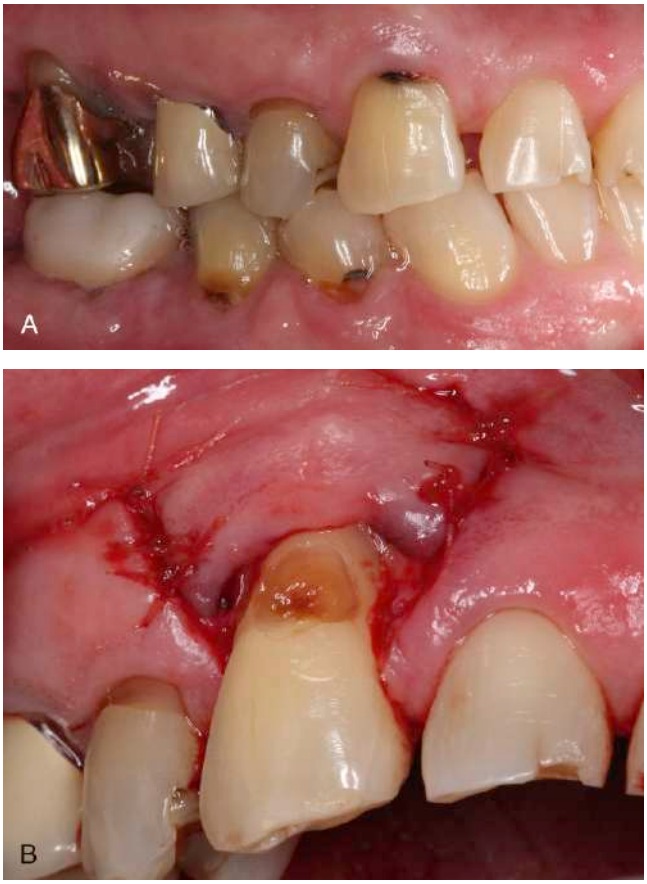


Figure 10.15 (A) Subgingival caries on teeth No. 6, No. 28 and No. 29. Ideally the cavitated areas need to be exposed surgically to allow for proper restoration. Further, a 3-mm distance for biologic width establishment is needed from the margin of the tooth preparation to the alveolar bone crest to promote best tissue health. (B) Tooth No. 6 with caries excavated, immediately after crown lengthening procedure.
(Courtesy: Dr Bruno Herrera.)

with evidence of osseous defect due to caries impinging the biologic width is present.

ii. Tooth Fracture

Crown lengthening is almost always necessary for subgingival crown-root fractures, to avoid continuous periodontal breakdown in the site. Subgingival fractures are normally already affecting biologic width considering their presentation. In the unusual situations where the biologic width is adequate after a subgingival fracture, it is very likely that crown lengthening will be necessary after tooth preparation for a crown, for instance, because the crown preparation is often placed more apically to where the fracture happened. When the fracture is not treated for several weeks or months, the periodontium and bone remodel on the site, moving away from the fracture margin, creating a bony ledge and a crater (moat-like defect) around the tooth. The area is likely to stay inflamed and become uncomfortable for the patient if not corrected.

iii. Inadequate Retention

Extensively compromised teeth due to extensive fractures and/or caries lesions often need to be altered prior to final restoration. When there is a need to create a ferrule in a tooth that will receive a post- and core-restoration, crown lengthening may be needed to facilitate its placement without violation of the biologic width and adequate retention form.

Crown lengthening may not be necessary if adequate ferrule can be achieved without violation of the biologic width. However, additional retention may be necessary; otherwise gingivectomy may be performed to temporarily remove gingival tissue.[68]

Gegauff highlighted that an attempt to gain adequate ferrule via crown lengthening procedure can negatively affect the tooth mechanical properties.[69] The apical relocation of the crown margin after crown lengthening procedures leads to a thinner cross section in the ferrule area. Further, there is a reduction in the crown-to-root ratio which also may negatively affect the outcome of the treatment. Orthodontic extrusion may be another option to expose tooth structure in some clinical situations of inadequate retention. Yet crown-to-root ratio and apical location of the ferrule also will occur in orthodontic extrusion.

Clinical Notes

There is an overall consensus that 1.5–2 mm of ferrule is necessary for adequate retention of crowns and protection of the integrity of the remaining tooth structures.

iv. Altered Passive Eruption

Crown lengthening or gingivectomy can also be indicated for esthetic reasons only or before restoring a tooth that has altered passive eruption. Altered passive eruption is the diagnosis when a significant amount of anatomic crown is covered by gingiva due to failure of proper apical migration of the soft tissue that covers the crown of the tooth. Often the patient displays a gummy smile with squared-looking teeth (Fig. 10.16A).[70] This condition affects esthetics and also increases plaque accumulation as the sulcus depth promotes easy apical plaque migration.[71] Gingivectomy may be performed when there is a distance of more than 3 mm from the CEJ to the alveolar crest, there is enough keratinized tissue and there is no evidence of bone defect (Fig. 10.16B and C). The location of the bone in relationship to the CEJ and the mucogingival junction are important parameters in treatment of altered passive eruption. They can be determined with bone sounding and clinical and radiograph examination. Gingivectomy is not advised when the band of keratinized tissue is insufficient (<3 mm).

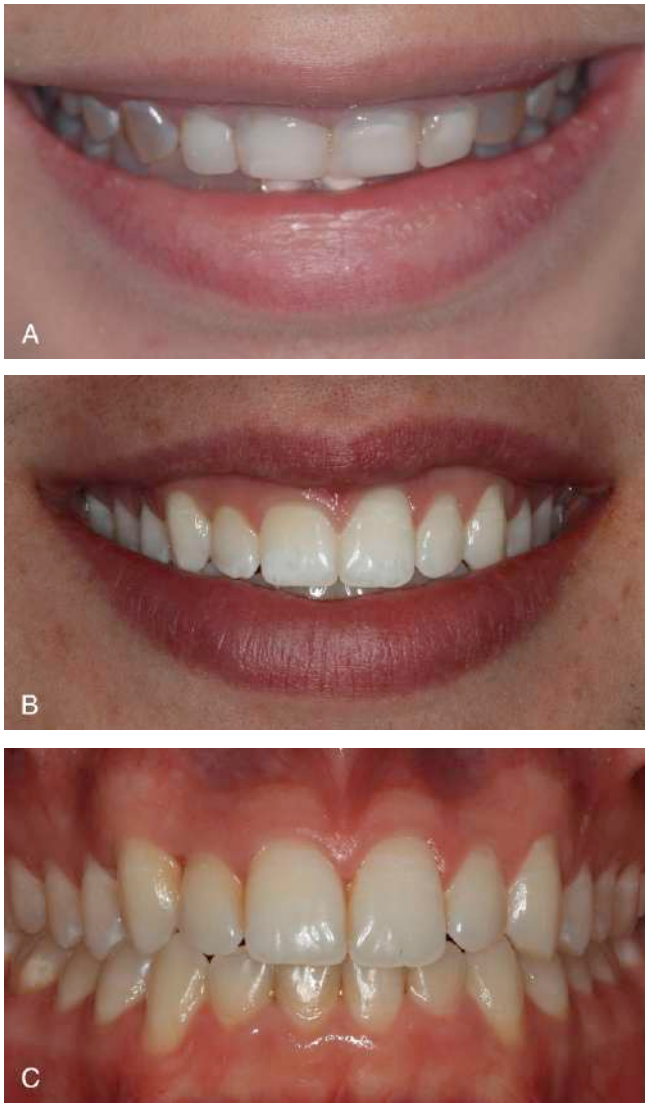


Figure 10.16 Altered passive eruption often results in a flat gingival contour and squared-shaped anterior teeth (A) or uneven gingival contours such as the case in *B*, where discrepancies can be noted between teeth No. 7 and No. 10 and teeth No. 8 and No. 9. case (B) with gingival discrepancy corrected by gingivectomy (C). ([A] Courtesy: Dr André Ritter; [B and C] Courtesy: Dr Akshay Kumarswamy.)

Distal Wedge

Distal wedge is commonly performed on the distal surface of molars (tuberosity or the retromolar pad) to facilitate hygiene, access and help eliminate periodontal pockets or facilitate access for proper restorative treatment (Fig. 10.17). The procedure is either complete removal of the tissue (if there is abundant amount of keratinized tissue even after removal) or partial ablation of tissue (to spare keratinized tissue). In any case, after tissue removal, soft-tissue coverage of the bone and more coronal tooth exposure is achieved. A direct restoration can be performed immediately after the surgical procedure provided adequate operating field isolation is achieved.



Figure 10.17 Tooth No. 2 with ill-fitting amalgam restoration at the subgingival distal margin. Distal wedge should have been performed prior to restoration to facilitate access for proper restorative treatment.

Effect of Restorative Treatment on the Periodontium

In the previous section, we reviewed the impact and indications of surgical procedures in restorative dentistry such as crown lengthening, gingivectomy and distal wedge. Often these procedures are crucial prior to proper restorative treatment. Conversely, it is also important to understand the impact of restorative procedures on periodontal health to prevent periodontal breakdown. It is critically important to understand the possible effects of restorative therapy (e.g. biologic width violation, materials, margins of restorations, provisional restorations, retraction cord trauma) on the periodontium.

Biologic Width Violation

Much has already been discussed in this chapter about the importance of biologic width preservation. Violation of the biologic width is often a result of caries removal and consequent preparation of the tooth for direct or indirect restorations leading to damage of the junctional epithelium and the connective tissue above the alveolar crest. The contact of the restoration margin with the junctional epithelium or the connective tissue below the gingival sulcus allows for bacterial colonization of the area. Recruitment of inflammatory cells and chronic inflammation of the site follow.[55] While polishing of the restoration may alleviate plaque accumulation and inflammation, there is extensive evidence of inflammation leading to bone resorption

in sites with biologic width violation. The inflammatory process is usually aggravated with indirect restorations as the marginal gap of a well-fitted crown is normally larger than that of a direct restoration. Sorensen and colleagues reported on marginal fit in relation to percentage bone loss and gingival flow and found positive correlation between marginal discrepancy of restorations and periodontal defects.[72] In summary, proper placement of the margin of the restoration (direct or indirect) in respect of the biologic width is critical.

Materials

Besides careful placement of restorations to avoid marginal overhangs, voids and rough restorations in close proximity to the periodontium, biocompatible materials should be favoured.

I. **Geristore**: Known for its biocompatibility, Geristore (DenMat, Lompoc, CA) is a dual-cure, hydrophilic resin-modified glass ionomer often recommended when the periodontium is extensively involved. Its manufacturer recommends it as liner, in direct pulp capping, in Class V and conservative Class I and Class II restorations, in root caries lesions, in subgingival restorations for fractured roots and resorption lesions, in root perforations and as retrograde filling material. Research has shown that Geristore has better or similar adhesion to tooth structures than some of the other glass ionomers and presents low microleakage.[77–80] Its biocompatibility has been extensively studied with mostly positive results being reported.[81–83] Geristore has shown to allow for better growth and cell morphology, and has improved toxicity profile of human gingival fibroblasts compared to Ketac Fil (3M Oral Care, St. Paul, MN) and IRM (Dentsply Sirona, York, PA).[82]

   Gupta and colleagues showed that Geristore had superior biocompatibility compared to MTA and glass ionomer when tested on human periodontal ligament cells.[81] A clinical study by Dragoo showed that Geristore was not different in terms of probing depth and gingival inflammation than other resin-based materials such as Dyract (Dentsply Sirona) and Photac Fil (3M Oral Care).[84] It was concluded that a material with biocompatibility, dual-cure capability, insolubility in oral fluids, low coefficient of thermal expansion, low polymerization shrinkage and low microleakage is ideal for subgingival restorations. Such properties are found in Geristore, which appears to be the preferred material among clinicians for cases where teeth are in need of subgingival restorations[85] (Fig. 10.18).

II. **Mineral trioxide aggregate (MTA)**: MTA appears to have a significantly improved tissue response and bone formation compared to composite resins, amalgams and glass ionomers.[73] The common response from other materials is the formation of fibrous connective tissue that will connect them to the alveolar bone. Despite this, MTA is not the material of choice for situations where the material is close to the gingiva, as it does not have the mechanical properties required to sustain occlusal stresses.[74] Further, a coronal restoration in contact with soft tissues would not need to promote bone growth but rather a healthy soft tissue response.

III. **Resin Modified Glass Ionomer Cement (RMGIC)**: Gomes and colleagues reported that the periodontal response to resin-modified glass ionomers was more favourable than the response to amalgam and composite resins in an animal study after 124 days.[75] In another more recent study, Sakallioglu and colleagues studied the release of inflammatory markers in gingival crevicular fluid in humans after restorations with metaloceramic, composite resin and amalgam restorations. Results showed a significant increase in certain inflammatory peptides compared to controls (enamel surface) for all groups with amalgam showing the highest levels after 4 weeks.[76]

Provisional Restorations and Restorative Margins

The margins of indirect restorations are often placed subgingivally for esthetic purposes, to provide preparation margin in tooth structure rather than in restorative material (restorative foundation) or for additional retention. However, even when placed at gingival level or slightly into the sulcus, some inflammatory tissue reactions will occur[86] (Fig. 10.19A). The presence of localized periodontal pathology has been found with indirect metaloceramic restorations independently of the supragingival or subgingival location of the margin. However, marginal adaptation is correlated with severity of inflammation.[86] To minimize the probability of tissue recession or chronic local inflammatory response, a well-adapted provisional restoration should always be placed and tissue reaction ideally observed for several weeks prior to final impression and placement of the final restoration. A tissue that is healthy after the provisional restoration is in place for several days or weeks is a good indication of no violation of the biologic width and that the periodontium will respond well to the final restoration margins (Fig. 10.19B).

Clinical Notes

A well-adapted provisional restoration also will facilitate gingival retraction and final impression and ensure that the tissue will rebound atraumatically (after impression).

When a poorly contoured provisional restoration is placed, the inflammatory process starts and can be perpetuated after the final restoration is cemented as the area remains injured.[87] It is important to correct margins and roughness of any provisional restoration as soon as

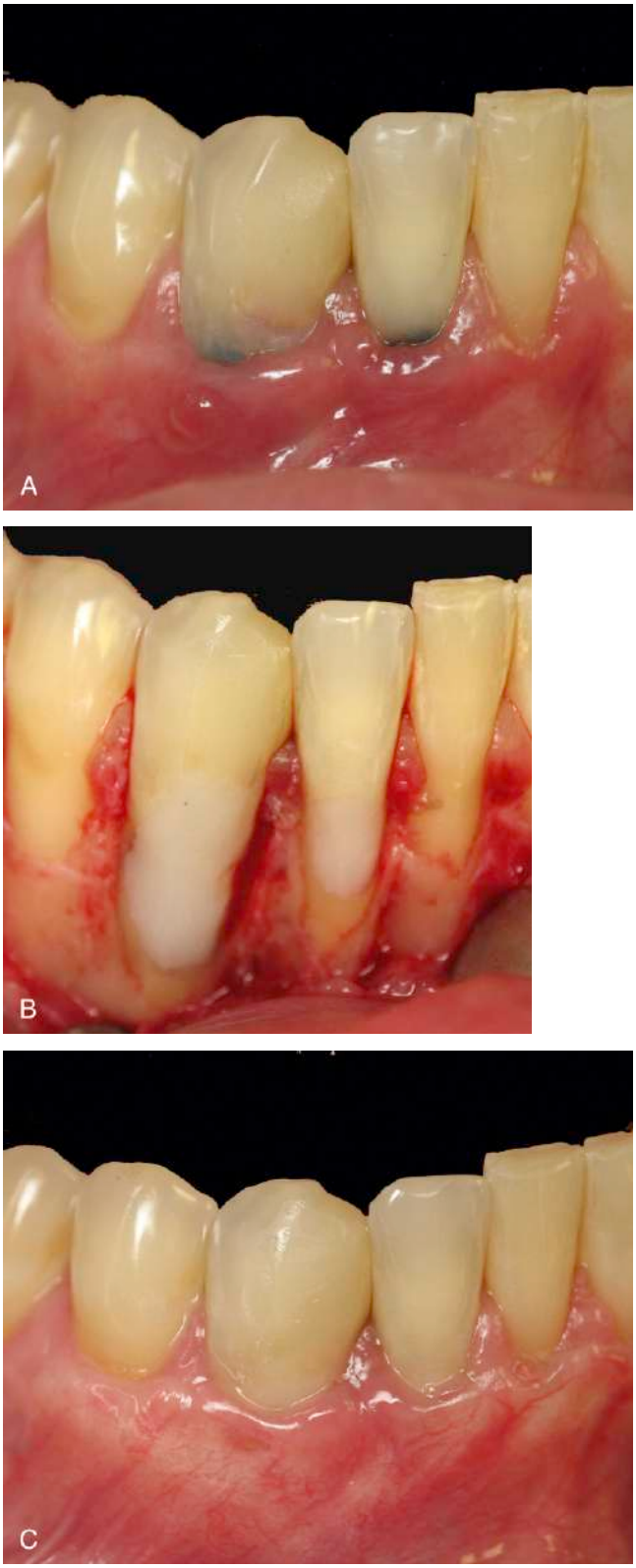


Figure 10.18 (A) Cervical caries lesion on teeth No. 26 and No. 27. (B) The lesions exposed and restored with Geristore. (C) Two-month follow-up with excellent tissue health.
(Courtesy: Dr Luis Perez.)



Figure 10.19 (A) Full-coverage restorations on teeth No. 8 and No. 9 with inadequate marginal preparation, crown adaptation and invasion of biologic width causing gingival inflammation. (B) Improved tissue health after crown lengthening to re-establish the biologic width and well-adapted temporary restoration.

detected to prevent and/or reverse damage leading to loss of periodontal attachment.

Further, placement of provisional restorations prior to proper initial healing after crown lengthening or gingivectomy can be problematic as it is not easy to determine the exact location where the alveolar crest and the biologic width will be re-established. Despite the periodontal procedure performed, the result may still be violation of the biologic width with development of an inflammatory lesion after surgery. When there is a need for immediate or early provisionalization after surgery, the restoration should rest on supragingival margins. This early provisionalization can help with the soft-tissue contouring and maintenance or papilla regrowth during the healing phase.[88] Further, some have advocated immediate provisionalization after crown lengthening surgery.[89] Such practice is only advised to extremely experienced periodontal-restorative teams and mostly indicated for patients with well-known positive tissue response to restorative procedures. Finally, to obtain a well-fitting provisional, it is important to understand that the tooth preparation/restoration margins need to be easily visualized and have enough intra-coronal depth to allow for thickness of material without overhanging margins.

Besides provisional restorations, any final indirect or direct restoration (even well fitting) harbours more plaque and bacteria than enamel and challenges the supportive periodontium.[90] Thus in the case of a poorly contoured restoration, periodontal breakdown and recurrent caries at the restoration–tooth interface is probable in the long term.

Retraction Cord and Impressions

When the tooth is receiving an indirect restoration, gingival displacement is usually needed to adequately capture

the preparation margins with either physical or digital impression methods.[91] This gingival displacement, or retraction, can be done mechanically, chemically, or surgically (electrosurgery or laser).

Retraction cords have been shown to cause a reversible gingival displacement that is greater than other types of materials (i.e. paste or foam).[92] Others have found that all retraction techniques cause temporary gingival displacement with inflammation with Ultrapak (Ultradent Products, Inc., South Jordan, UT) cord without impregnation of haemostatic agent causing most bleeding but fastest recovery compared to Magic FoamCord (Coltène, Altstätten, Switzerland) and Expasyl (Kerr Corporation, Orange, CA).[93] In this study, all retraction techniques caused an acute injury after 1 day of retraction, which took 1 week to heal in the Ultrapak and the Magic FoamCord groups. The Expasyl group had the highest gingival index of inflammation and showed the slowest healing. Its use may also cause tooth sensitivity. In another study, the Magic FoamCord system was a less traumatic method of gingival retraction compared to cords in slightly subgingival preparations. However, the material was less effective than the single cord retraction technique in preparations with margins deeper than 2 mm.[94]

When evaluating gingival retraction cords versus electrosurgery, Wostmann and colleagues did not find any significant difference in the quality of the marginal fit of the crowns as long as the impression technique utilized two (light and heavy) impression materials.[95] To date there is no comparison between laser, electrosurgery, cord and cordless materials in regard to gingival health and quality of impression. It is currently accepted that retraction systems cause some temporary damage to the integrity of the gingival tissues; however, the soft-gingival tissue heals in a few days provided there is adequate marginal adaptation of the temporary restoration.

Summary

When planning restorations, thorough evaluation of periodontal health, impact of restorative treatment on periodontium and patient perceptions of esthetics and understanding of oral health is warranted. Respecting the periodontium boundaries while restoring tooth structures is vital for the success of the restorative treatment and the overall oral homoeostasis. Ill-fitting restorations and undiagnosed periodontal defects in teeth that will be restored can complicate treatment and lead to further disturbance of periodontal health. Although periodontal disease is primarily caused by dental plaque, there are modifying and significant alterations caused by restorative treatment that contribute to the establishment and progression of periodontal breakdown.

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